Symmetric Hadamard matrices of orders 532 and 732

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Abstract

We construct symmetric Hadamard matrices of orders $532 = 4 \cdot 133$ and $732 = 4 \cdot 183$. To our knowledge these are the first symmetric Hadamard matrices of these orders: $v = 133$ and $v = 183$ appear in the most recent published list of odd $v < 200$ for which no symmetric Hadamard matrix of order $4v$ was known. Both come from the propus construction applied to cyclic difference families whose blocks are unions of orbits of a multiplier subgroup: parameters $(133; 61, 64, 64, 57; 113)$ with the subgroup $\langle 26 \rangle$ of order 6 in $\mathbb{Z}_{133}^*$, and $(183; 81, 91, 91, 83; 163)$ with the subgroup $\langle 34 \rangle$ of order 5 in $\mathbb{Z}_{183}^*$. The second family was found with a compression-guided exhaustive search exploiting $183 = 3 \cdot 61$. Both existence statements are formally verified in Lean 4 with Mathlib: we prove the propus construction in general and check the required autocorrelation identities by kernel evaluation. We also report (computational, not formally verified) negative evidence for the smallest open order $260 = 4 \cdot 65$: no cyclic propus difference family in $\mathbb{Z}_{65}$ admits a nontrivial multiplier.

1 Introduction

A Hadamard matrix of order n is an $n \times n$ matrix H with entries ± 1 and $HH^{\mathsf{T}} = nI_n$. It is *symmetric* if $H = H^{\mathsf{T}}$. Symmetric Hadamard matrices of order $4v$ are conjectured to exist for every $v \geq 1$; as a symmetric Hadamard matrix of order n yields one of order $2n$, the essential case is v odd. For many years the main source of new orders has been the *propus construction* of Seberry and Balonin [1], in the cyclic form developed by Balonin, Đoković and collaborators [3, 4, 5, 6, 7]: a Goethals–Seidel difference family (X_1, X_2, X_3, X_4) in $\mathbb{Z}_v$ with $X_2 = X_3$ and X_1 symmetric gives a symmetric Hadamard matrix of order $4v$.

The Handbook of Combinatorial Designs (Craigen–Kharaghani, 2007, Table 1.52) lists the odd $v < 200$ for which no symmetric Hadamard matrix of order $4v$ was known. Using the propus construction, many of these cases were settled ($v = 23, 29, 39, 43, 47, 59, 67, 73, 103, 109, 113, 127, 151, 191$, see [2, 3, 4, 5, 6]); Đoković [6, §3] records that the list of undecided cases reduces to

$$v = 65,\ 89,\ 93,\ 101,\ 107,\ 119,\ 133,\ 149,\ 153,\ 163,\ 167,\ 179,\ 183,\ 189,\ 193.$$

The unresolved list of Seberry and Balonin [1, §4] also contains 133 and 183. Subsequently $v = 93$ was settled in [7] and $v = 189$ follows from the orders $q(q+1)$, $q \equiv 3 \pmod 8$ a prime power ($q = 27$), of [8]. We found no later work treating $v = 133$ or $v = 183$ (we checked the arXiv listing of Đoković's papers up to October 13, 2026, the propus catalogue of Balonin, a scan of recent arXiv papers with "Hadamard" in the title, and web searches); the classical constructions do not

apply: $531 = 3^2 \cdot 59$, $265 = 5 \cdot 53$, $731 = 17 \cdot 43$ and $365 = 5 \cdot 73$ are not prime powers, neither 532 nor 732 is of the form $q(q+1)$, and $532 = 4 \cdot 7 \cdot 19$, $732 = 4 \cdot 3 \cdot 61$ admit no Kronecker factorisation into orders of symmetric Hadamard matrices.

Theorem 1.1. *There exist symmetric Hadamard matrices of orders* 532 *and* 732*.*

The difference families behind Theorem 1.1 are given in Sections 3 and 4; they were found by exhaustive meet-in-the-middle searches over unions of multiplier orbits. Section 5 describes the Lean formalisation. In Section 6 we report what our searches say about the smallest open order 260.

2 The propus construction

For $x, y : \mathbb{Z}_v \to \mathbb{Z}$ put $C_{x,y}(t) = \sum_{u \in \mathbb{Z}_v} x(u)\, y(u+t)$; the periodic autocorrelation of x is $\mathrm{PAF}_x(t) = C_{x,x}(t)$. For $X \subseteq \mathbb{Z}_v$ let χ_X be the ± 1 sequence equal to -1 on X and $+1$ off X. Subsets $X_1, \dots, X_4$ of sizes k_i form a *Goethals–Seidel difference family* if $\sum_i \mathrm{PAF}_{\chi_{X_i}}(t) = 0$ for all $t \neq 0$; equivalently they form a difference family with $\lambda = \sum k_i - v$.

Proposition 2.1 (propus construction [1, 3])**.** *Let $a, b, d : \mathbb{Z}_v \to \{\pm 1\}$ satisfy $\mathrm{PAF}_a(t) + 2\mathrm{PAF}_b(t) + \mathrm{PAF}_d(t) = 0$ for all $t \neq 0$. Let A, B, D be the circulant matrices $A_{ik} = a(k-i)$ etc., and $R_{ik} = [i + k = 0]$. Then*

$$P = \begin{pmatrix} -A & BR & BR & DR \\ BR & RD & A & -RB \\ BR & A & -RD & RB \\ DR & -RB & RB & A \end{pmatrix}$$

satisfies $PP^{\mathsf{T}} = 4vI$. If moreover $a(-u) = a(u)$ for all u, then $P = P^{\mathsf{T}}$.

Proof. Index rows and columns by $\{0,1,2,3\} \times \mathbb{Z}_v$. The (i,k) entries of A, AR, RA are $a(k-i)$, $a(-i-k)$, $a(i+k)$. Each block of PP^{T} is a sum of four terms $\sum_k x(\cdot)y(\cdot)$, and the substitutions $u = k - i$, $u = -i - k$, $u = i + k$ turn each into $\pm C_{x,y}(\pm i \pm j)$ or $\pm K_{x,y}(\pm i \pm j)$ with $K_{x,y}(t) = \sum_u x(u)y(t-u)$. The diagonal blocks equal $C_{a,a}(s) + 2C_{b,b}(s) + C_{d,d}(s)$ with $s = \pm(i-j)$ (using $C_{x,x}(-s) = C_{x,x}(s)$), which is $4v[i = j]$ by hypothesis and $C_{x,x}(0) = v$. In each off-diagonal block the terms cancel in pairs by $K_{x,y} = K_{y,x}$ and $C_{x,y}(t) = C_{y,x}(-t)$. Symmetry: AR, RA are symmetric for every circulant, the blocks in positions $(1,2), (1,3), (2,1), (3,1)$ all equal BR, and A (in positions $(1,1), (2,3), (3,2), (4,4)$) is symmetric iff a is. All of this is formalised in Lean (Section 5). □

3 The construction for $v = 133$

Let $H_6 = \langle 26 \rangle = \{1, 11, 20, 26, 87, 121\} \leq \mathbb{Z}_{133}^*$; note $26 \equiv 5 \pmod 7$ has order 6 and $26 \equiv 7 \pmod{19}$ has order 3, and $-1 \notin H_6$. The H_6-orbits on $\mathbb{Z}_{133}$ are $\{0\}$, six orbits of size 3 (on $7\mathbb{Z}_{133} \setminus \{0\}$), one orbit of size 6 (on $19\mathbb{Z}_{133} \setminus \{0\}$) and 18 further orbits of size 6. For $S \subseteq \mathbb{Z}_{133}$ write $H_6 S = \bigcup_{s \in S} H_6 s$. Put

$$\begin{aligned} X_1 &= H_6\{0,1,2,3,4,5,12,19,24,28,29,70\}, && |X_1| = 61, \\ X_2 = X_3 &= H_6\{0,2,4,5,14,16,28,29,30,35,45,56,59,70\}, && |X_2| = 64, \\ X_4 &= H_6\{1,2,6,12,16,19,29,35,56,59,70\}, && |X_4| = 57. \end{aligned}$$

Explicitly,

$$\begin{aligned}
X_1 = \{&0,1,2,3,4,5,11,12,19,20,22,24,26,28,29,33,36,38,40,41,42,44,\\
&46,48,51,52,53,55,57,60,63,70,73,76,78,80,81,82,85,87,89,91,92,93,\\
&95,97,100,104,105,107,109,111,113,114,121,122,128,129,130,131,132\},\\
X_2 = \{&0,2,4,5,14,16,17,21,22,28,29,30,35,36,39,40,41,42,43,44,45,48,\\
&51,52,53,54,55,56,58,59,62,63,64,68,70,71,73,74,79,80,82,83,84,85,\\
&89,90,91,96,98,100,102,104,105,106,109,112,115,116,117,119,125,126,129,130\},\\
X_4 = \{&1,2,6,11,12,16,17,19,20,22,23,26,29,35,38,40,41,43,46,48,51,52,\\
&53,54,56,57,59,61,62,66,70,71,74,76,79,84,87,89,90,91,95,105,107,109,\\
&112,113,114,116,117,119,120,121,122,123,126,129,132\}.
\end{aligned}$$

X_1 is symmetric ($X_1 = -X_1$). With $a = \chi_{X_1}$, $b = \chi_{X_2}$, $d = \chi_{X_4}$ one checks

$$\mathrm{PAF}_a(t) + 2\mathrm{PAF}_b(t) + \mathrm{PAF}_d(t) = 0 \qquad (t = 1, \ldots, 132), \tag{3.1}$$

so (X_1, X_2, X_2, X_4) is a cyclic difference family with parameters $(133; 61, 64, 64, 57; 113)$ ($61 + 2 \cdot 64 + 57 - 133 = 113$, and $61 \cdot 60 + 2 \cdot 64 \cdot 63 + 57 \cdot 56 = 113 \cdot 132$). By Proposition 2.1 the propus matrix is a symmetric Hadamard matrix of order 532; this proves Theorem 1.1. Because all blocks are H_6-invariant, (3.1) needs to be checked only on the 13 orbits of $\langle H_6, -1\rangle$ on $\mathbb{Z}_{133} \setminus \{0\}$, but our verifications check all 132 shifts.

How the family was found. For a subgroup $H \leq \mathbb{Z}_v^*$ we search for propus families whose blocks X_2, X_4 are unions of H-orbits and X_1 is a union of $\langle H, -1\rangle$-orbits (the standard multiplier ansatz of Đoković et al.). Our program enumerates all candidates for X_4 of the prescribed size whose power spectral density satisfies $\mathrm{PSD} \leq 4v$ at every frequency, stores them in a hash table keyed by the vector of autocorrelations on the $\langle H, -1\rangle$-classes of shifts, enumerates candidates for X_1 ($\mathrm{PSD} \leq 4v$) and X_2 ($\mathrm{PSD} \leq 2v$), and looks up $-(\mathrm{PAF}_a + 2\mathrm{PAF}_b)$ for every pair with $\mathrm{PSD}_a + 2\mathrm{PSD}_b \leq 4v$. For $H = \langle 26\rangle$ the 26-orbit space is exhausted for all 20 propus parameter sets of $v = 133$ in about a minute; only $(133; 61, 64, 64, 57; 113)$ admits solutions, namely 72 families, which fall into 4 orbits of size 18 under the multiplier group $\mathbb{Z}_{133}^*$ (all listed in the data file). As a check, the same program re-finds structured families of the kind reported in [6], e.g. for $v = 57$, $H = \langle 7\rangle$ and parameters $(57; 27, 26, 26, 22; 44)$ it finds 864 (ordered) families, and it reproduces independently computed counts for small test cases ($v = 35, 45, 73$).

4 The construction for $v = 183$

Let $H_5 = \langle 34\rangle = \{1, 34, 58, 70, 142\} \leq \mathbb{Z}_{183}^*$; here $34 \equiv 1 \pmod 3$ and 34 has order 5 modulo 61, so H_5 acts only on the $\mathbb{Z}_{61}$-coordinate of $\mathbb{Z}_{183} \cong \mathbb{Z}_3 \times \mathbb{Z}_{61}$. Its orbits are the three fixed points $0, 61, 122$ and 36 orbits of size 5. Put

$$\begin{aligned}
X_1 &= H_5\{0,4,6,7,15,17,18,23,24,28,39,46,47,56,59,69,91\}, && |X_1| = 81,\\
X_2 = X_3 &= H_5\{0,4,6,7,8,9,10,13,14,15,17,20,23,36,39,43,46,47,59\}, && |X_2| = 91,\\
X_4 &= H_5\{0,1,2,3,4,5,6,9,10,15,23,25,46,47,52,59,61,78,122\}, && |X_4| = 83.
\end{aligned}$$

Explicitly,

$X_1 = \{0, 4, 6, 7, 15, 17, 18, 19, 21, 23, 24, 28, 29, 33, 35, 37, 39, 40, 45, 46, 47, 48,$
$49, 50, 53, 54, 55, 56, 59, 63, 66, 69, 71, 72, 74, 77, 79, 83, 84, 86, 91, 92, 97, 99,$
$100, 104, 106, 109, 111, 112, 114, 117, 120, 124, 127, 128, 129, 130, 133, 134, 135, 136, 137, 138, 143, 144,$
$146, 148, 150, 154, 155, 159, 160, 162, 164, 165, 166, 168, 176, 177, 179\},$

$X_2 = \{0, 4, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17, 19, 20, 21, 22, 23, 29, 31, 35, 36,$
$38, 39, 40, 43, 45, 46, 47, 48, 49, 50, 53, 54, 55, 59, 62, 65, 66, 67, 71, 75, 76, 79,$
$80, 81, 82, 86, 89, 92, 95, 97, 98, 100, 104, 106, 109, 110, 115, 117, 119, 120, 123, 124, 126, 127,$
$128, 131, 134, 135, 136, 138, 139, 141, 143, 144, 146, 151, 155, 156, 157, 158, 164, 165, 168, 171, 176, 178,$
$179, 180, 181\},$

$X_4 = \{0, 1, 2, 3, 4, 5, 6, 9, 10, 15, 19, 21, 23, 25, 27, 31, 34, 46, 47, 49, 50, 52,$
$53, 54, 58, 59, 60, 61, 64, 68, 70, 73, 78, 81, 86, 88, 90, 96, 97, 100, 101, 102, 103, 104,$
$106, 107, 109, 116, 117, 118, 120, 121, 122, 123, 127, 128, 132, 134, 135, 136, 138, 139, 140, 142, 143, 144,$
$146, 151, 153, 155, 156, 157, 161, 163, 164, 165, 167, 169, 170, 174, 176, 179, 180\}.$

Again $X_1 = -X_1$, and with $a = \chi_{X_1}$, $b = \chi_{X_2}$, $d = \chi_{X_4}$ we have $\mathrm{PAF}_a(t) + 2\mathrm{PAF}_b(t) + \mathrm{PAF}_d(t) = 0$ for $t = 1, \ldots, 182$; the parameters are $(183; 81, 91, 91, 83; 163)$ ($81 + 182 + 83 - 183 = 163$, $81 \cdot 80 + 2 \cdot 91 \cdot 90 + 83 \cdot 82 = 163 \cdot 182$). Proposition 2.1 gives a symmetric Hadamard matrix of order 732.

How the family was found. The H_5-invariant space has 39 orbits per block, too many for the plain search of Section 3. We write a block as an assignment of a *column pattern* $P_O \in \{0,1\}^3$ to each H_5-orbit O of $\mathbb{Z}_{61}$ (the entries (x, y), $y \in O$, are -1 iff the x-th bit of P_O is set). The 61-compression $c_y = \sum_x \chi_X(x, y)$ of a propus family must itself satisfy the compressed autocorrelation identity on $\mathbb{Z}_{61}$ with total energy $4v = 732$; we first enumerate all compressed triples by meet-in-the-middle (4^{13} candidates per block), and then uncompress only compatible ones ($3^{\#\{O : |c_O| = 1\}}$ choices each), computing the power spectral density incrementally per column pattern and matching the full autocorrelation vectors through a sorted table. For the parameter set $(183; 81, 91, 91, 83; 163)$ the search finds exactly 432 H_5-invariant propus families (about 13 minutes on one core, 6 GB), forming 18 orbits of size 24 under the multiplier group $\mathbb{Z}_{183}^*$; all 432 were verified independently. All other eleven propus parameter sets of $v = 183$ were exhausted without H_5-invariant solutions. The program reproduces the counts 864 and 216 of the plain search for $v = 57$, $H = \langle 7 \rangle$.

5 Formal verification

The file `lean/Main.lean` (Lean 4.35.0, Mathlib) contains three parts; below, Unicode symbols are rendered in ASCII (`forall`, `->`, `\/`, `Int`; a `*` between a scalar and `1` stands for scalar multiplication $\bullet$).

Part 1 (general). Definitions `C` (periodic cross-correlation), `K` (periodic convolution) and `propus a b d : Matrix (Fin 4 × ZMod v) (Fin 4 × ZMod v) ℤ`, and the theorems

```
theorem propus_mul_transpose (a b d : ZMod v -> Int)
    (ha1 : forall u, a u = 1 \/ a u = -1) (hb1 : forall u, b u = 1 \/ b u = -1)
    (hd1 : forall u, d u = 1 \/ d u = -1)
    (hpaf : forall t : ZMod v, t != 0 -> C a a t + 2 * C b b t + C d d t = 0) :
```

```
    propus a b d * (propus a b d).transpose = (4 * (v : Int)) * 1
theorem propus_transpose (a b d : ZMod v -> Int) (ha : forall u, a (-u) = a u) :
    (propus a b d).transpose = propus a b d
```

proved by reindexing finite sums (sixteen block identities).

Parts 2 and 3 (the instances). For $v = 133$ and $v = 183$ the three blocks are encoded as natural-number bitmasks; the identity (3.1) (for $v = 133$ as `∀ t : ZMod 133, t ≠ 0 → C sA sA t + 2 * C sB sB t + C sD sD t = 0`) and the symmetry of X_1 are proved by `decide +kernel`, i.e. by evaluation in the Lean kernel (no `native_decide`); likewise for $v = 183$. The main statements are

```
theorem Sym532.symmetric_hadamard_532 :
    exists H : Matrix (Fin 532) (Fin 532) Int,
      (forall i j, H i j = 1 \/ H i j = -1) /\ H.transpose = H /\
      H * H.transpose = (532 : Int) * (1 : Matrix (Fin 532) (Fin 532) Int)
```

and the identical statement `Sym732.symmetric_hadamard_732` with 532 replaced by 732; the witnesses are the propus matrices reindexed along `finProdFinEquiv : Fin 4 × Fin 133 ≃ Fin 532` (resp. `Fin 4 × Fin 183 ≃ Fin 732`). *Statement audit:* the three conjuncts say exactly that H is a ± 1 matrix, symmetric, and Hadamard of order 532 (resp. 732). `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` (see `lean/lean_output.txt`). Independently, a Python/NumPy script builds all 72 matrices of order 532 and all 432 matrices of order 732, and checks $H = H^{\mathsf{T}}$ and $HH^{\mathsf{T}} = 4vI$ in exact integer arithmetic.

6 The smallest open order 260

The smallest order for which no symmetric Hadamard matrix is known is $260 = 4 \cdot 65$ [5, 7]. For $v = 65$ the propus parameter sets are exactly $(65; 27, 31, 31, 27; 51)$, $(65; 31, 27, 27, 31; 51)$, $(65; 28, 29, 29, 28; 49)$ and $(65; 29, 28, 28, 29; 49)$. We call $h \in \mathbb{Z}_{65}^*$ a *multiplier* of a propus family if $hX_i = X_i + c_i$ for all blocks. Every nontrivial subgroup of $\mathbb{Z}_{65}^* \cong \mathbb{Z}_4 \times \mathbb{Z}_{12}$ contains 16 (generating the unique subgroup of order 3) or one of the involutions $14, 51, 64$, and for each of these h one checks that invariance up to translation can be turned into exact invariance by translating X_2, X_4 (the symmetric block X_1 is then automatically exactly invariant). We searched exhaustively, for all four parameter sets, the spaces of families whose blocks are unions of orbits of H:

H	orbits on $\mathbb{Z}_{65}$	method / result
every subgroup of order ≥ 3	≤ 25	full meet-in-the-middle: none
$\langle 51 \rangle$, $(x, y) \mapsto (x, -y)$ on $\mathbb{Z}_5 \times \mathbb{Z}_{13}$	35	MITM with 5- and 13-compression filters: none
$\langle 64 \rangle = \{\pm 1\}$	33	same: none
$\langle 14 \rangle$, $(x, y) \mapsto (-x, y)$	39	row decomposition, compression filters: none

Proposition 6.1 (computational)**.** *No cyclic propus difference family in $\mathbb{Z}_{65}$ admits a nontrivial multiplier.*

This explains why the multiplier ansatz, which produced the orders $4v$ for $v = 73, 93, 113, 127, 191$ [3, 6, 7] and $v = 133, 183$ here, cannot reach $v = 65$. It is also in sharp contrast with a Gaussian heuristic for the number of solutions, which predicts of the order of 10^2–10^3 families in the $\langle 16 \rangle$-

and $\langle 51\rangle$-invariant spaces and $\sim 10^4$ in the $\langle 14\rangle$-invariant space (the same heuristic is accurate for $v = 57, 73$ and for the $\langle 26\rangle$-space of $v = 133$). These statements are computer results and are not formally verified; the programs were validated by reproducing known counts on smaller cases. Local search (tabu search, simulated annealing) in the full space finds propus families quickly for $v \leq 37$ but not for $v = 65$. Whether a propus family without multipliers exists for $v = 65$ remains open.

7 Limitations

Novelty of the orders 532 and 732 rests on a literature search (Đoković's lists and later papers up to 2025, web search); we cannot exclude an unpublished construction. The negative statements about $v = 65$ (Section 6) depend on our search programs (validated on known cases) and are not certified. The order 260 remains open.

References

[1] J. Seberry, N. A. Balonin, Two infinite families of symmetric Hadamard matrices, *Australas. J. Combin.* 69(3) (2017) 349–357.

[2] O. Di Matteo, D. Ž. Đoković, I. S. Kotsireas, Symmetric Hadamard matrices of order 116 and 172 exist, *Spec. Matrices* 3 (2015) 227–234; arXiv:1503.04226.

[3] N. A. Balonin, D. Ž. Đoković, D. A. Karbovskiy, Construction of symmetric Hadamard matrices of order $4v$ for $v = 47, 73, 113$, *Spec. Matrices* 6 (2018) 11–22; arXiv:1710.03037.

[4] N. A. Balonin, D. Ž. Đoković, Symmetric Hadamard matrices of orders 268, 412, 436 and 604, *Inf. Control Syst.* 2018 no. 4; arXiv:1803.08787.

[5] L. V. Abuzin, N. A. Balonin, D. Ž. Đoković, I. S. Kotsireas, Hadamard matrices from Goethals–Seidel difference families with a repeated block, *Inf. Control Syst.* 2019 no. 5, 2–9.

[6] D. Ž. Đoković, Some new symmetric Hadamard matrices, *Inf. Control Syst.* 2022 no. 2; arXiv:2101.05429.

[7] D. Ž. Đoković, Hadamard matrices: skew of orders 276, 292 and symmetric 372, arXiv:2301.02751 (v3, 2024).

[8] D. Ž. Đoković, Orthogonal designs and symmetric Hadamard matrices, arXiv:2508.17141 (2025).

[9] R. Craigen, H. Kharaghani, Hadamard matrices and Hadamard designs, in: Handbook of Combinatorial Designs, 2nd ed., CRC Press, 2007, 273–280.